Khoa Nguyen - Lead Data Engineer

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Lead Data Engineer and former Technical Lead with 10+ years of experience building and owning end-to-end data platforms from zero to business-critical scale. Proven track record delivering self-service analytics (80% adoption), powering low-latency customer-facing analytics (20% of product features), and driving 30%+ marketing impact through trusted metrics and scalable data architecture.

Regularly write technical articles on Medium, sharing guides on software engineering, data-related topics, and personal experiences from real-world projects.

CORE SKILLS

Data Engineering & Architecture

- ETL / ELT pipeline design (Python-based)
- Event-based & behavioral data modeling
- Streaming + batch hybrid architectures
- Data quality, validation, and monitoring

Analytics Engineering

- Dimensional modeling (fact & dimension tables)
- Metric definition & semantic layer
- dbt-based analytics engineering
- Self-service BI enablement

Data Platforms & Warehousing

- Amazon Redshift
- BigQuery
- Snowflake
- ClickHouse
- PostgreSQL, MySQL
- Performance tuning & query optimization
- Storage, partitioning, retention strategies

Processing & Compute

- Apache Spark
- AWS Glue, Athena
- High-volume batch & near real-time processing
- Low-latency analytics pipelines

Technical Leadership

- End-to-end platform ownership
- Architecture & technical decision-making
- Mentoring engineers & analysts
- Cross-functional collaboration (Product, Marketing, Finance)

Data Orchestration & Transformation

- SQLMesh, DBT
- Dagster, Airflow

PROFESSIONAL & ENGLISH SKILLS

- Fluent English; extensive experience in international working environments
- Team leadership, mentoring, and technical alignment
- Effective collaboration in multicultural and distributed teams
- Strong stakeholder communication and technical facilitation

PROFESSIONAL EXPERIENCES

Lead Data Engineer, Flodesk

Mar 2020 — Dec 2025

Flodesk is a design-focused email marketing platform enabling small businesses to grow through **email-driven marketing**.

- Acted as the technical lead and end-to-end owner of the analytics and data platform, driving architectural decisions, technical roadmap, and cross-functional alignment.
- Designed and evolved a company-wide data warehouse using modular, model-driven transformation workflows (dbt-based, SQLMesh-aligned patterns) with strict idempotency discipline and re-runnable backfill strategies.

- Architected Medallion-style data layers (raw → curated → business-ready) to ensure scalable transformations and consistent metric definitions across teams.
- Built and maintained **scalable pipelines** processing **100B+ email and user events**, supporting both batch and near real-time workloads.
- Designed partition-aware merge strategies, workload isolation, and cost-aware query optimization aligned with modern cloud warehouse architectures (Redshift, Snowflake POC, BigQuery-style patterns).
- Implemented rigorous data validation processes including row-count reconciliation, metric diffing, and controlled parallel runs prior to production rollout.
- Enabled **self-service analytics** adoption (**~80% active usage**) through standardized semantic models and BI-ready datasets.
- Powered **~20% of customer-facing analytics features** using low-latency, high-availability data systems.
- Improved **marketing effectiveness by 30%+** through unified reporting, accurate attribution, and trusted metrics.
- Led and mentored a cross-functional data team, enforcing clean SQL standards, modular design principles, and PR-based engineering workflows.
- Conducted a Snowflake proof-of-concept to evaluate performance isolation, scalability, and long-term warehouse evolution strategy.

Technical Lead / Software Designer, Kamora Limited Company

Feb 2019 — Feb 2020

- Acted as Technical Lead & key contributor in delivering end-to-end product features across backend, frontend, and mobile platforms.
- Led system design and architecture decisions for multiple startup products targeting European markets, ensuring scalability and production readiness.
- Took ownership of core modules, mentored team members, and drove technical alignment in fast-paced, early-stage environments.
- Ensured stable releases and rapid iteration cycles, balancing speed and quality for client-facing European projects.

Technical Lead / Software Designer, Sioux High Tech Software

Aug 2013 — Jan 2019

- Served as Technical Lead / key member in the design and delivery of large-scale, mission-critical systems for European enterprise and government clients.
- Led end-to-end system design, from requirements analysis to architecture definition and production deployment.
- Worked directly with European customers on technical requirements, architectural trade-offs, and long-term system evolution.
- Played a central role in delivering systems with high reliability, safety, and compliance standards.
- Collaborated and led within multinational engineering teams, maintaining high technical quality and best practices across distributed teams.

HIGHLIGHTED PROJECTS

Company-wide Data Platform & Single Source of Truth, United States

- Built a unified data foundation consolidating **product, marketing, and finance data** into a consistent analytics layer.
- Established standardized datasets and metrics to ensure **data consistency and trust** across business functions.
- Enabled organization-wide analytics adoption, supporting **self-service reporting for ~80% of frequent users**.

High-volume Event Analytics for Email Marketing, United States

- Designed a **high-throughput event ingestion and analytics system** handling **100B+ email and user events**.
- Supported both **near real-time and batch analytics** for data exploration and customer-facing use cases.
- Provided a reliable analytics backbone for **~20% of customer-facing product features**.

Self-service Analytics Enablement, United States

- Developed analytics models and BI-ready datasets to support **self-service analysis across business teams**.
- Reduced reliance on ad-hoc data requests by enabling users to independently explore and answer business questions.
- Improved overall **data literacy and analytics efficiency** across the organization.

Smart Car, The Netherlands

- Contributed to a **government-backed initiative in the Netherlands** aimed at reducing traffic congestion on **highway A58**.
- Built a system to **collect and process real-time vehicle telemetry data via USB** from in-car devices.
- Analyzed driving patterns and traffic conditions to generate **driver-facing instructions and speed recommendations**.

- Supported improved traffic flow through **real-time, data-driven guidance** delivered on mobile devices.

ACHIEVEMENTS

- Microsoft Certified Xamarin Developer
- Speaker at Vietnam Mobile Day

EDUCATION

Bachelor of Information Technology (Excellent)

Da Nang University of Technology, 2008 — 2013

Additional Information

Hobbies: Football, Writing technical articles